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Environment

Low-intensity prescribed fire has no immediate impact on soil physical and chemical properties or enzymatic activity in the Post Oak Savanna ecoregion, Texas, USA

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Abstract

The historic development of the Post Oak Savanna ecoregion in Texas, USA, was shaped by natural wildfires. However, wildfire suppression has caused an increase in encroaching woody species. To address this issue, land management strategies incorporate prescribed fire (PF) for the control of encroaching species. Wildfire has been shown to impact soil physical and hydraulic properties as well as soil enzymatic activities; however, the effects of PFs, which are typically less intense than wildfires, are not as well-described. This study sought to quantify the effects, if any, of PF on soil physical and hydraulic properties and enzymatic activities in the Post Oak Savanna ecoregion. Comparative analysis was conducted on soil samples from 0 to 5 cm and 5 to 10 cm measured prior to and immediately after PF. Soil wet aggregate stability, unsaturated and saturated hydraulic conductivities, water retention, organic carbon, and enzyme activities were quantified. Results indicate that neither soil properties nor enzymatic activities were significantly altered due to low-intensity PF. Low-intensity PF has no immediate effect on soil physical and chemical properties or soil enzyme activity. Low-intensity PF is an effective land management tool that has no impact on the soil properties measured here.

Abbreviations: ANOVA, analysis of variance; BG, β -glucosidase; ENRTA, Ecology and Nature Resource Teaching Area; NAG, N-acetyl- β -D-glucosaminidase; NRCS, Natural Resources Conservation Service; PF, prescribed fire; PNP, *p*-nitrophenol; SOC, soil organic carbon; THAM, tris(hydroxymethyl)aminomethane; VWC, volumetric water content at time of sampling; WDPT, water droplet penetration time.

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Plain Language Summary

The historic development of the Post Oak Savanna ecoregion in Texas, USA, was shaped by natural wildfires. However, wildfire suppression has caused an increase in woody species that would have historically been controlled by fire. To address this issue, land managers often use intentionally set and controlled fires, called prescribed fires (PFs), to control woody plants. The effects of PFs, which are typically less intense than wildfires, are not well-described in the literature. The goal of this work was to measure the effects, if any, of PF on soil properties and enzymatic activities in the Post Oak Savanna ecoregion. Comparative analysis was conducted on soil samples collected from the surface and subsurface prior to and immediately after PF. Results indicate that neither soil properties nor enzymatic activities were significantly altered due to low-intensity PF.

1 | INTRODUCTION

Prescribed fires (PFs), or controlled burns, are intended to mimic the role of naturally occurring wildfire in an ecosystem. PFs are closely monitored, performed under safe weather conditions, and last only for a short duration. PFs have been shown to reduce the incidence and intensity of natural wildfires by reducing the amount of available fuel and managing invasive species, which often contribute excess fuel (DiTomaso & Johnson, 2006; Hodgson, 1968; Wu et al., 2023). Studies have shown that the effect PFs have on soil properties varies depending on fire type, seasonality, and ecosystem (Alcañiz et al., 2018). PFs typically affect only the top centimeter of topsoil (Vega et al., 2014) but are capable of affecting soils up to a depth of 15 cm (Ulery et al., 2017). Depending on the frequency and intensity of burns, PFs can facilitate soil carbon (C) and nitrogen (N) cycling, increase interactions between various plant species and soil microbes, and influence soil moisture regimes (Alcañiz et al., 2018; Landry, 1998; P. Smith et al., 2016; Teague & Dowhower, 2022). Shifts in plant populations due to PFs may also affect the hydrologic properties of the soils of these ecosystems (Cawson et al., 2016; Huxman et al., 2005).

Previous studies have presented mixed results regarding the impacts of PF on soils. Some suggest that PF can improve vegetation cover and subsequently decrease runoff and soil erosion (Garza & Blackburn, 1985), but a number of studies have shown that PF can cause hydrophobicity, leading to decreases in infiltration and increased runoff (Alcañiz et al., 2018; Meeuwig, 1969; Morales et al., 2000). Landry (1998) concluded that seasonal fluctuations in precipitation, vegetative growth, and temperature influence hydrological responses, and that these processes, coupled with PF, can significantly influence soil physical properties. Supporting this, Robichaud (2000) observed a general trend of increased hydrophobicity and increased runoff in soil under a conifer

forest after PF; however, the effect of fire on soil hydraulic properties depended on its severity, or the physical impact of the fire on vegetation and soil, and its intensity, or the amount of energy released during the fire. For example, it was shown that a low-intensity, low-severity burn where 65%–100% of vegetation remained had no significant impact on infiltration and runoff rates, whereas a high-intensity, high-severity burn did (Robichaud, 2000). One prior study conducted in the Post Oak Savanna ecoregion in Texas demonstrated an increase in sediment loss and a decrease in infiltration rate after a PF (Lloyd-Reilly et al., 1984).

PF can also impact soil enzymatic activities (Zhou et al., 2022), which can further alter soil properties. Soil enzymatic activities affect organic matter decomposition and thus soil nutrient availability, soil water retention, and soil hydrophobicity (Bhattacharyya & Furtak, 2022). High temperatures from PFs can denature or destroy soil enzymes (Zhou et al., 2022) or reduce microbial biomass (Dooley & Treseder, 2012), leading to impaired organic matter decomposition and soil nutrient availability (Mekonnen et al., 2019). This reduction in soil nutrient availability could further reduce microbial activity, which, paired with an increase in organic matter build-up, could increase soil hydrophobicity and decrease soil water retention (Allton et al., 2007). Activities of β -glucosidase (BG), an enzyme involved in cellulose decomposition, have been shown to either decrease (Boerner & Brinkman, 2003; Eivazi & Bayan, 1996) or remain unaffected (Boerner et al., 2000) following PFs, with decreases in activity potentially linked to repeated burns versus a single PF event. Alternatively, N-acetyl- β -D-glucosaminidase (NAG), an important enzyme in the coupled carbon and nitrogen cycles, and in degrading the compound chitin, the main constituent of fungal cell walls (Deng & Popova, 2011), was shown to increase for months following a PF, potentially due to an increase in chitin availability or chitinolytic organisms (Boerner et al., 2000). Other recent papers have investigated

the impact of fire on NAG activity, given its significance in C and N cycling and the known impact of fire on the dynamics of those two cycles (Gutknecht et al., 2010; O'Kelley et al., 2024; Xu et al., 2024).

The impact of PFs is dependent on specific site characteristics, climate, and vegetation. In areas that were historically shaped by fires but no longer have them unless prescribed, such as the Texas Post Oak Savanna ecoregion, understanding the effects of PFs on soil is essential for proper land management. Prior to the suppression of natural fires by European settlers beginning around 1850, the mean fire return interval in this region was 5.9 years; however, fire frequency decreased significantly after that time, leading to an increase in woody plants, and currently little is known regarding the frequency and severity of PFs in this region (Sparks et al., 2012).

The objective of this study was to quantify the effects of a PF event on soil physical and chemical properties and enzymatic activities in the Post Oak Savanna ecoregion in south-central Texas. We had a threefold hypothesis: the PF would not significantly alter the soil physical properties at any depth, PF would alter the surface soil chemical properties and enzymatic activities, and PF would not significantly affect subsurface soil chemical properties or microbial activity. These hypotheses were informed by prior studies that have shown that PF does not impact soil physical properties (Pierson et al., 2008; Tan et al., 1986), that PF may impact chemical properties or enzymatic activity near the soil surface (Scharenbroch et al., 2012), but that PF does not significantly impact chemical properties or enzymatic activity in the subsurface (Scharenbroch et al., 2012). We tested this hypothesis by performing a comparative analysis of soil physical and chemical properties and enzymatic activity before and after PF in a Post Oak Savanna.

2 | METHODS

2.1 | Site description

The study site is located at the Ecology and Nature Resource Teaching Area (ENRTA) in Brazos County, Texas, within the Post Oak Savanna ecoregion. The ENRTA is a biological field station managed by Texas A&M University's Department of Rangeland, Wildlife, and Fisheries Management, in collaboration with the Department of Ecology and Conservation Biology. The ENRTA has a total area of 382 ha, a mean elevation of 91 m a.s.l., a mean annual rainfall (1991–2020) of 1060 mm, and a mean annual air temperature (1991–2020) of 20.8°C (deVilleneuve et al., 2023; National Oceanic & Atmospheric Administration, 2022). The study site is characterized as a Post Oak Savanna grassland (hereafter referred to as “grassland”) with Boonville fine sandy loam (Fine, smectitic, thermic Chromic Vertic Albaqualfs) soils according to the Natural Resources Conservation Service (NRCS) Web

Soil Survey database (Soil Survey Staff, 2023). The vegetation at the study site consisted of native grassland vegetation (*Schizachyrium scoparium* and *Bothriochloa laguroides*) with sparse Post oak (*Quercus stellata*) trees. This site was burned in November 2019 and May 2021, and was scheduled to be burned again in spring 2023 (at the time of this study). To our knowledge, the site had not been burned prior to 2018. From the early-to-mid-1900s until 2018, the site was an unmanaged fallow pasture grazed by livestock; prior to that, the site is believed to have been used for cotton production beginning in the late 1800s.

2.2 | Prescribed fire

While the PF occurred throughout the entire area shown in Figure 1, our study focused on the southernmost portion, the first of three subsections within this area. The delineation of these subsections may be seen as breaks in the vegetation in Figure 1. The PF occurred on the afternoon of March 3, 2023, and was a low-intensity burn covering the entirety of the grassland treatment area (Figure 1). Burn severity was not measured directly, though experts present during the burn as well as visual evidence of the fire (Figure 2) support this claim (Dr. Joseph Veldman, personal communication, 2025). The fuel load prior to the burn was estimated to be 1000 kg ha⁻¹ based on visual evaluations performed by the fire ecologist who oversaw the burn (Dr. Joseph Veldman, personal communication, 2025). Fire weather at the time of the burn was as follows: winds at 7 mph NW, temperature of 22°C, and relative humidity of 43%. Based on this information, the dead fine fuel moisture is estimated to have been in the range of 7%–11%. The mean flame length was estimated to be 1.5 m based on observations and photos of the burn, and the burn rate was estimated to be low based on the total burn time (1 h) and the total burned area (2.6 ha).

2.3 | Soil sampling

To capture the effects of PF on soils in the Post Oak Savanna ecoregion, we collected soil samples once prior to and once immediately after a PF. Preburn samples were collected on March 2, 2023, and postburn samples were collected the following day after the PF. The pre- and postburn samples were not paired samples and were not collected from the same locations within the field, with a mean distance of 13 m from any preburn to any postburn sample.

Three sets of undisturbed soil samples (100 cm³) were collected from depths of 0 to 5 cm and 5 to 10 cm from representative points within each treatment using Eijkelkamp bulk density sampling rings (Royal Eijkelkamp). At the same time and location as undisturbed soil samples were collected, unsaturated hydraulic conductivity (*K*) was also measured

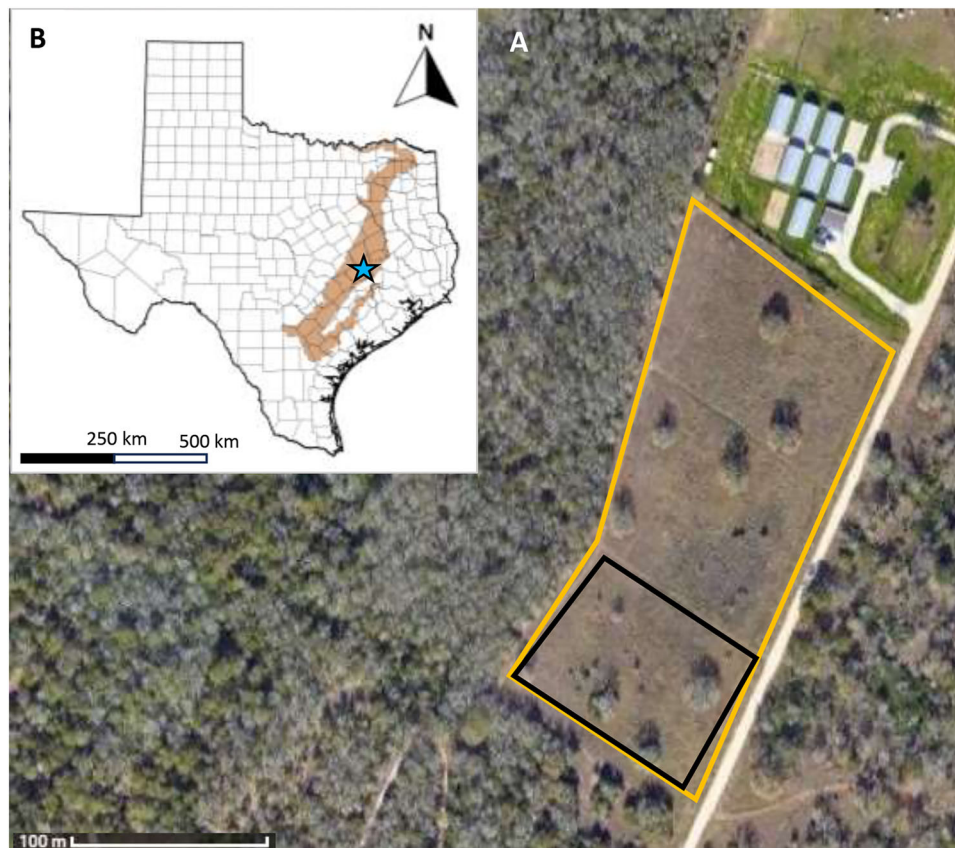


FIGURE 1 (A) Prescribed fire location shown in yellow outline and soil sampling area shown in black outline. (B) Map of Texas with the Post Oak Savanna ecoregion shaded in brown and Brazos County, TX, marked with blue star. North arrow applies to (A) and (B).

using a tension infiltrometer (METER Group). Additionally, six disturbed soil samples were collected using a sterilized 1.25-cm-diameter push probe adjacent to undisturbed samples for analysis of soil organic carbon (SOC) and enzyme activity. Between sampling campaigns, the site received 13.5 mm of rainfall overnight, which potentially affected measurements and soil samples collected on the day of the PF.

2.4 | Soil texture, bulk density, and water content at sampling

The percentages of sand, silt, and clay in the soil samples were determined using the hydrometer method of particle size analysis described by Grossman and Reinsch (2002). To account for the high level of variability in the sand readings, two hydrometer readings were taken, and their means were used for the sand calculation. For clay percentage calculations, another reading was taken in a specific time interval based on the temperature of the solution (~7 h). The silt percentage was determined by subtracting the sum of the values calculated for sand and clay from 100. A check sample was analyzed in all particle analyses to ensure the accuracy and consistency of the results, and subsamples were weighed, oven-dried, and

reweighed to account for residual water in soil samples at the time of analysis.

Intact soil cores were used to estimate the bulk density of the soil by dividing the mass of the soil material after drying at 105°C for 24 h by the volume of the sampling rings. Immediately upon returning the samples to the laboratory, their masses were measured and used to estimate the volumetric water content at the time of sampling.

2.5 | Unsaturated hydraulic conductivity

Unsaturated hydraulic conductivity (K) was measured adjacent to the three soil sampling locations within each treatment using a tension infiltrometer with a suction of -2 cm according to the method proposed by Zhang (1997). Unsaturated hydraulic conductivity was calculated by the following equation:

$$K = \frac{C_1}{A} \quad (1)$$

where C_1 is the slope of the curve of the cumulative infiltration versus the square root of time, and A is a value relating the



FIGURE 2 Grassland area during the low-intensity prescribed burning.

van Genuchten (1980) soil hydraulic parameters for a given soil type to the suction and radius of the infiltrometer disk. Here, an A value of 3.9099 was used based on the classification of soil at our site as a sandy loam (see Table 1), as well as a 2.25-cm infiltrometer radius and an applied suction of -2 cm.

2.6 | Hydrophobicity

Soil hydrophobicity was tested in the field according to the method described by Dekker et al. (2009) where the water droplet penetration time (WDPT) was measured at different depths on the disturbed soil cores prior to bagging them (similar to fig. 1 in Dekker et al. [2009]). Disturbed soil samples were extracted from the 0- to 10-cm depth near the locations of unsaturated hydraulic conductivity measurements within each treatment using a push probe sampler with a 1.25 cm diameter. With the soil still in the soil probe, WDPT at the 0- and 5-cm depths was measured using one droplet of water at each depth. The time taken for the water droplets to infiltrate into the soil sample was measured using a stopwatch and recorded.

2.7 | Saturated hydraulic conductivity measurements

The saturated hydraulic conductivity (K_s) of intact soil cores was measured using the constant head method with an Eijkelkamp laboratory permeameter (Royal Eijkelkamp). The permeameter basin was filled with a 5 mmol CaCl_2 solution to prevent soil dispersion. The soil saturated hydraulic conductivity (K_s) was calculated from the following equation:

$$K_s = \frac{V \times L}{A \times t \times h} \quad (2)$$

where V is the volume of water (cm^3), L is the length of the soil samples (cm), A is the cross-sectional area of the sample (cm^2), t is the length of time (s), and h is the hydraulic head (cm).

2.8 | Water retention at -33 kPa (θ_{-33})

Soil water retention at -33 kPa can be used as an estimation of field capacity and was measured using a Tempe cell (Soil Moisture Equipment Corporation) analysis as described by Dane and Hopmans (2002). Prior to analysis, eighteen 1-bar ceramic plates were saturated in a 5 mmol CaCl_2 solution for 24 h or until saturation was achieved. Samples were then pressurized to 33 kPa and checked daily until the sample mass difference between readings was less than 1 g. Once equilibrium was achieved, the final mass was recorded.

2.9 | Water retention at -1500 kPa (θ_{-1500})

Soil water retention at -1500 kPa was measured using a pressure chamber (Soil Moisture Equipment Corporation) according to Dane and Hopmans (2002). After equilibrium was reached, the mass of the soil samples was recorded. Then, samples were oven-dried at 105°C for 24 h, after which their dry weight was also recorded. The volumetric soil water content was found by multiplying the gravimetric water content by the bulk density of the sample.

TABLE 1 Mean and standard error (in parentheses) values of sand, silt, and clay percentages.

Variable	Units	Depth	Preburn	Postburn
Sand	%	0–5	66.5 ($\pm$ 1.54)	67.4 ($\pm$ 1.27)
		5–10	67.4 ($\pm$ 0.75)	67.6 ($\pm$ 2.95)
Silt	%	0–5	28.5 ($\pm$ 1.12)	27.6 ($\pm$ 0.02)
		5–10	25.1 ($\pm$ 0.02)	28.2 ($\pm$ 2.57)
Clay	%	0–5	5.02 ($\pm$ 0.73)	5.02 ($\pm$ 1.26)
		5–10	7.52 ($\pm$ 0.73)	4.18 ($\pm$ 0.42)
ρ_b	g cm ⁻³	0–5	1.22 ($\pm$ 0.04)*	1.25 ($\pm$ 0.05)*
		5–10	1.52 ($\pm$ 0.03)	1.45 ($\pm$ 0.03)
K	cm day ⁻¹	0	147 ($\pm$ 45.6)	141 ($\pm$ 15.1)
VWC	cm ³ cm ⁻³	0–5	0.43 ($\pm$ 0.06)	0.45 ($\pm$ 0.05)
		5–10	0.40 ($\pm$ 0.04)	0.42 ($\pm$ 0.02)
WDPT	s	0–5	0.00 ($\pm$ 0.00)	0.22 ($\pm$ 0.11)
		5–10	0.00 ($\pm$ 0.00)	0.11 ($\pm$ 0.11)
K_s	cm day ⁻¹	0–5	8.67 ($\pm$ 0.14)*	8.94 ($\pm$ 0.29)*
		5–10	8.34 ($\pm$ 0.13)	8.25 ($\pm$ 0.07)
θ_{-33}	cm ³ cm ⁻³	0–5	0.19 ($\pm$ 0.02)	0.17 ($\pm$ 0.03)
		5–10	0.18 ($\pm$ 0.01)	0.15 ($\pm$ 0.02)
θ_{-1500}	cm ³ cm ⁻³	0–5	0.04 ($\pm$ 0.01)	0.04 ($\pm$ 0.00)
		5–10	0.03 ($\pm$ 0.00)	0.03 ($\pm$ 0.00)
AS	%	0–5	85.8 ($\pm$ 6.56)	94.5 ($\pm$ 0.47)
		5–10	80.5 ($\pm$ 15.8)	87.6 ($\pm$ 1.27)
SOC	g kg ⁻¹	0–5	19.5 ($\pm$ 2.60)*	17.4 ($\pm$ 3.97)*
		5–10	10.2 ($\pm$ 2.67)	8.18 ($\pm$ 0.23)
N	g kg ⁻¹	0–5	2.07 ($\pm$ 0.25)*	2.24 ($\pm$ 0.45)*
		5–10	1.20 ($\pm$ 0.26)	1.35 ($\pm$ 0.36)
BG	mg PNP kg ⁻¹ h ⁻¹	0–5	197 ($\pm$ 27.7)*	304 ($\pm$ 98.3)*
		5–10	0.00 ($\pm$ 0.00)	29.8 ($\pm$ 26.6)
NAG	mg PNP kg ⁻¹ h ⁻¹	0–5	68.6 ($\pm$ 11.1)*	58.6 ($\pm$ 9.11)*
		5–10	37.5 ($\pm$ 2.24)	19.3 ($\pm$ 3.29)

Note: There were no significant differences between treatments for any variable.

Abbreviations: θ_{-1500} , volumetric water content at -1500 kPa; θ_{-33} , volumetric water content at -33 kPa; ρ_b , bulk density; AS, wet aggregate stability; BG, β -glucosidase; K , unsaturated hydraulic conductivity; K_s , saturated hydraulic conductivity; N, total nitrogen; NAG, N-acetyl- β -D-glucosaminidase; PNP, p -nitrophenol; SOC, soil organic carbon; VWC, volumetric water content at time of sampling; WDPT, water droplet penetration time.

*Statistically significant differences between depths ($\alpha = 0.05$).

2.10 | Aggregate stability

Aggregate stability was determined for all soil samples using a wet sieving apparatus (Royal Eijkelkamp) following the methods described by Kemper and Rosenau (1986). Four grams of 1- to 2-mm-sized oven-dried soil aggregates were moistened with deionized water by fine mist and agitated for 3 min in a 5 mmol CaCl₂ solution. The soil that passed through the 60-mesh screen after 3 min was considered the unstable fraction. To determine the stable fraction of aggregates, the remaining soil was agitated in a sodium hexametaphosphate dispersant solution until all soil particles passed through the 60-mesh screen. The unstable aggregates that passed easily

through a 60-mesh sieve and the stable aggregates that did not pass easily were weighed separately. The proportion of stable aggregates was then determined by dividing the weight of stable aggregates by the combined weight of stable and unstable aggregates, with the result expressed as a percentage.

2.11 | Soil organic carbon and nitrogen

The method for measuring inorganic C content of the samples was adapted from Sherrod et al. (2002). For the generation of the standard curve, 2 g of eight standards (0.00, 0.24, 0.30, 0.60, 1.2, 2.4, 12, and 18 g kg⁻¹) were measured into serum bottles along with 2 g of each soil sample. Hydrochloric

ric acid (HCl), 6 M, with 3% ferrous chloride ($\text{FeCl}_2 \bullet 4\text{H}_2\text{O}$) was made by diluting concentrated HCl with deionized water to a 1:1 ratio and allowing it to cool before adding 3 g of FeCl_2 for every 100 g of HCl. Four gelatin capsules containing 1 mL of reagent each were added to each serum bottle immediately upon filling, and the bottles were sealed. The samples were mixed gently and allowed to cool for 6 h before taking readings. Voltage readings were taken using a pressure calcimeter by inserting the probe needle of the pressure calcimeter into the lid of each bottle (Sherrod et al. 2002). Once the readings were stable, the results from the pressure calcimeter were recorded, and the percentage of inorganic carbon was calculated using the steps given in Sherrod et al. (2002).

For measuring total carbon and nitrogen, samples were air-dried and ground to pass through a 2-mm sieve. Twenty $\text{mg} \pm 0.05$ mg of soil were weighed and placed into tin capsules, the tops of which were folded to prevent loss of any soil particles. Total C and N concentrations were measured via total combustion using a Vario EL Cube Elementar elemental analyzer (Elementar Americas Inc.). To obtain organic C totals for each soil sample, inorganic C values were subtracted from total C values.

2.12 | Soil enzyme activities

Potential soil enzyme activities of β -glucosidase (BG) and N-acetyl- β -D-glucosaminidase (NAG) were determined via a scaled-down version of the spectrophotometric method described by Stott (2019) in 2.2-mL 96-well plates. Reagent concentrations were adjusted based on Acosta-Martínez et al. (2019) and Dick (2020). Field moist samples were air-dried, ground, and sieved (≤ 2 mm). Then each sample was divided into triplicate samples. For BG, assay samples were incubated for 1 h at 37°C in a pH 6.0 modified universal buffer and 0.0625 mol of *p*-nitrophenol (PNP) β -D-glucopyranoside substrate. For NAG, assay samples were incubated for 1 h at 37°C in pH 5.5 sodium acetic acid and 0.0125 mol of PNP N-acetyl- β -D-glucosaminide substrate. Each sample was assayed in triplicate with two negative controls to which substrates were added after the incubation. After incubation, a stop solution consisting of 2.5 M CaCl_2 and 12.0 pH, 0.5 M tris(hydroxymethyl)aminomethane (THAM) buffer was added to all wells. For both enzyme assays, the well plates were sealed and shaken for 5 min before incubation. After stopping the reaction, the plates were centrifuged for 5 min at 75 RCF, 175 μL of the supernatant was transferred to a microplate, and, in the case of BG, 40 μL of the supernatant product was diluted with 135 μL of THAM diluent. Enzyme activities were determined colorimetrically at 405 nm in a microplate reader. Control samples' absorbances were subtracted from those of the soil samples. The final

enzyme activities were expressed as $\text{mg PNP kg soil}^{-1} \text{ h}^{-1}$.

2.13 | Statistical analysis

All statistical analyses were performed using MATLAB (MathWorks). All measured soil properties were analyzed using multi-way analysis of variance (ANOVA) to test the effects of treatment and depth, as well as treatment $\times$ depth interactions using $\alpha = 0.05$. Additionally, to assess the relationships between variables, we calculated the Pearson correlation coefficients (r) and their corresponding p -values.

3 | RESULTS AND DISCUSSION

Our findings are summarized in Table 1, which shows the mean values of all analyzed parameters and their statistical significance, and in Figure 3, which shows a correlation plot of all measured variables in this study. We discuss each variable below, as well as the correlations between that variable and others that were measured. Table S1 includes p -values used to determine differences between treatments, differences between depths, and treatment $\times$ depth interactions.

3.1 | Soil texture

We found that PF had no influence on soil particle size distribution, which was expected and aligns with prior studies that show low-intensity burns do not affect soil texture (Scharenbroch et al., 2012). The dominant soil textural class was sandy loam, which was expected based on NRCS Soil Survey classifications at the site. There were no significant differences ($p > 0.05$) in sand, silt, or clay percentages between pre- and postburn samples or between depths, which was expected. Similarly, there were no treatment $\times$ depth interaction effects—the effect of the treatment on texture was not influenced by the depth and vice versa. Variations in the impact of PF on soil texture have been observed across different studies. While some studies have reported alterations in soil texture resulting from PF, those changes are typically due to erosion and deposition of soil particles within the landscape, resulting from the loss of protective cover. The lack of significant effects of burning on soil particle size distribution is expected for low-intensity PFs (Alcañiz et al., 2018; Ketterings et al., 2000).

Across all treatments, sand content was significantly negatively correlated with silt ($r = -0.76$, $p = 0.004$, Figure 3). This was expected, since percentages of silt and clay must decrease if the sand content of the soil increases. Sand content was also significantly positively cor-

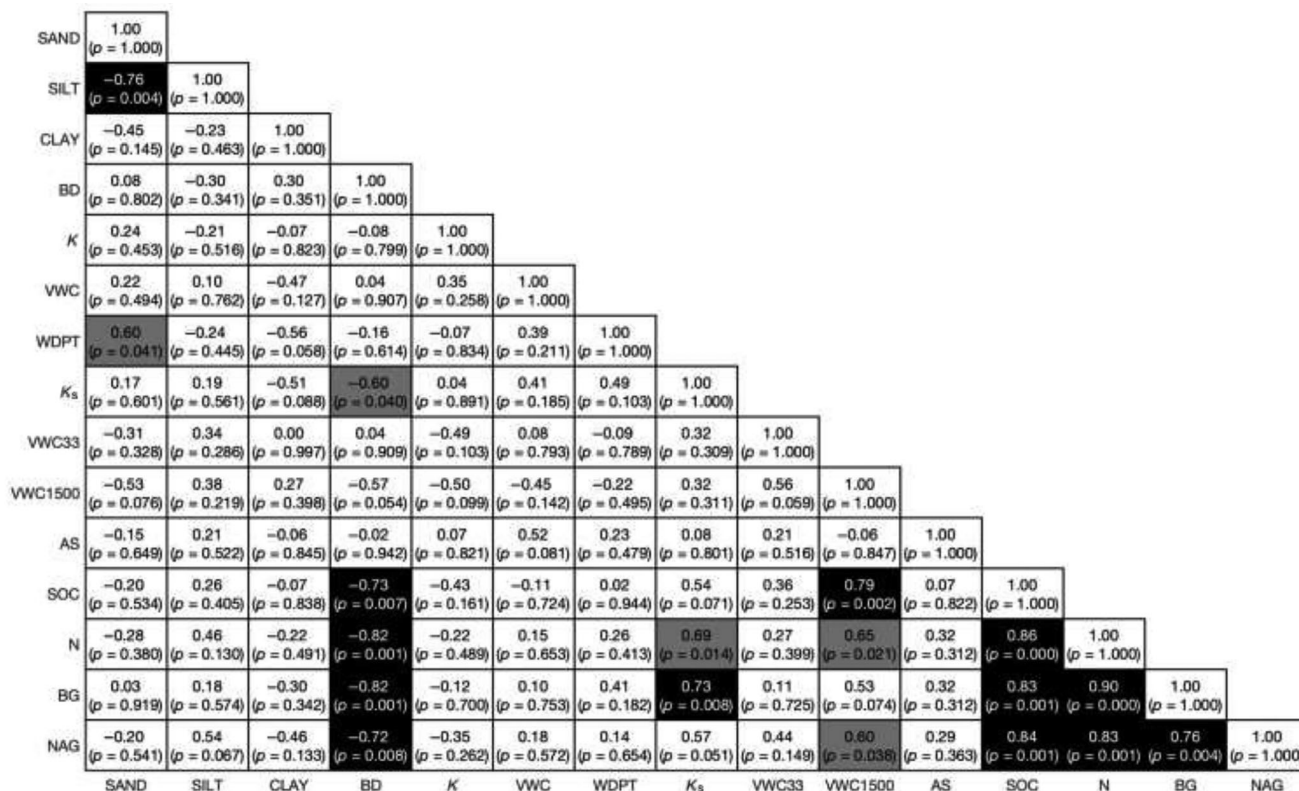


FIGURE 3 Correlation matrix of analysis results across all sites and depths, with $p < 0.01$ in black, $0.1 < p < 0.05$ in gray, and $p > 0.05$ in white. SAND is percent sand, SILT is percent silt, CLAY is percent clay, BD is bulk density, K is unsaturated hydraulic conductivity, VWC is the volumetric water content at the time of sampling, WDPT is water droplet penetration time, K_s is saturated hydraulic conductivity, VWC33 is volumetric water content at -33 kPa, VWC1500 is volumetric water content at -1500 kPa, AS is wet aggregate stability, SOC is soil organic carbon, N is total nitrogen, BG is soil enzyme activity of β -glucosidase, and NAG is enzyme activity of N-acetyl-glucosaminidase.

related with the WDPT ($r = 0.60$, $p = 0.041$), which aligns with prior findings that certain types of lipids found in sandy soils can influence water repellency (Jiménez-Morillo et al., 2017; Mao et al., 2015). No other significant correlations were found between soil texture and any other variables.

3.2 | Bulk density

Mean bulk density values ranged from 1.22 g cm^{-3} in the 0- to 5-cm depth for the grassland preburn treatment to 1.52 g cm^{-3} at the 5- to 10-cm depth in the grassland preburn (Table 1). Statistical analyses indicate that significant differences exist in bulk density between the two sampling depths ($p = 0.000$), with higher values at the 5- to 10-cm depth in both treatments. However, there were no differences in bulk density among treatments and no treatment $\times$ depth interaction effects. Thus, bulk density was found to be unaffected by the burning on the grassland, which has also been observed in prior studies (Ozslan Parlak et al., 2015; Schacht et al., 1996; Sherman et al., 2005) and was expected.

Bulk density was significantly negatively correlated with K_s ($r = -0.60$, $p = 0.040$), SOC ($r = -0.73$, $p = 0.007$), total N ($r = -0.82$, $p = 0.001$), and BG ($r = -0.82$, $p = 0.001$) and NAG ($r = -0.72$, $p = 0.008$) enzyme activities. The negative correlation between bulk density and K_s was expected since an increase in bulk density inevitably leads to decreases in pore space through which water may flow (Rawls et al., 1998; Schroeder et al., 2022). Similarly, the negative relationships between bulk density and SOC and total N were also expected since an increase in organic matter has been shown to decrease bulk density (Ruehlmann & Korschens, 2009). The negative relationships between bulk density and BG and NAG enzyme activities align with the findings of other studies (Kujur et al., 2012; Li et al., 2002; Udawatta et al., 2009).

3.3 | Unsaturated hydraulic conductivity and water content at time of sampling

Because of the positive relationship between soil water content and K , we have included these results together here. Mean values of K , which were measured only at the surface, and

VWC (volumetric water content at time of sampling) for each treatment are shown in Table 1. Unsaturated hydraulic conductivity values ranged from 63.8 to 221 cm day⁻¹ in the grassland preburn treatment and from 110 to 159 cm day⁻¹ in the grassland postburn treatment (data not shown), indicating a high level of variability in K prior to the burn. However, there were no significant differences found between treatments, indicating that PF does not affect K values. Across all treatments, K values were not significantly correlated with any other variable.

Surface (0–5 cm) volumetric soil moisture values at the time of sampling are higher than subsurface values in both treatments. Additionally, soil moisture levels at both depths are higher in the grassland postburn treatment than those in the preburn treatment (Table 1). This is a result of the overnight rainfall event prior to the PF, which confounds the observed relationship between soil water content and PF effects. However, the near-saturated conditions in the grassland prior to the PF ($\overline{\text{VWC}} = 0.43 \text{ cm}^3 \text{ cm}^{-3}$) suggest that the rainfall event likely had little effect on soil moisture. No significant differences were found in soil moisture values between treatments, nor were there any treatment $\times$ depth interaction effects, further suggesting that low-intensity PF has little effect on soil moisture (Table 1). Soil moisture at the time of sampling was not found to be significantly correlated to any other variable (Figure 3). Overall, our results indicate that the PF did not significantly affect the K value or soil moisture levels in the grassland. These findings align with those of previous studies, which also found that unsaturated hydraulic conductivity and soil water content are not affected by low-intensity PF (Meira Castro et al. 2014; Plaza-Álvarez et al., 2019).

3.4 | Hydrophobicity

Water droplet penetration time (WDPT) readings were all below 5 s and fell under Class 0 when classified quantitatively using the index proposed by Dekker and Jungerius (1990), indicating no sampling site experienced hydrophobicity before or after the PF. The maximum WDPT was 0 s for the grassland preburn and 1 s for the grassland postburn (data not shown, means shown in Table 1), but no significant differences were found between treatments or between depths, and there were no treatment $\times$ depth interaction effects. Surprisingly, we did not see any differences in WDPT between the preburn and postburn grassland treatments, which is contrary to what has been observed in many previous studies (Agbeshie et al., 2022; Everett et al., 1995). This is potentially due to the rainfall event received the evening before the prescribed burning, which led to wet soil conditions that would act to buffer surface soil temperatures during the fire and sub-

sequently decrease the likelihood of hydrophobicity on the soil surface (Badía et al., 2017; De Jonge et al., 1999; Miyata et al., 2007). Across all sites, WDPT was significantly correlated only with sand content ($r = 0.60$, $p = 0.041$, Figure 3). Some prior work has shown that sandy soils are more likely to be hydrophobic than more finely textured soils (Blackwell, 1993; Giovannini & Lucchesi, 1983), but hydrophobicity has been demonstrated in soils of all textural classes (Doerr et al., 2006, 2000).

3.5 | Saturated hydraulic conductivity (K_s)

ANOVA for K_s shows significant differences between depths ($p = 0.045$), but no significant differences between treatments and no treatment $\times$ depth interaction effects, indicating that the PF did not significantly change the K_s (Table 1). These results were expected, given the low intensity of the PF considered here, and follow those of Plaza-Álvarez et al. (2019), who also found no significant difference between K_s before and after a PF. Across treatments, K_s values were significantly correlated with BG enzyme activity ($r = 0.73$, $p = 0.008$), total N ($r = 0.69$, $p = 0.014$), and bulk density ($r = -0.60$, $p = 0.040$).

3.6 | Water retention at -33 kPa (θ_{-33}) and -1500 kPa (θ_{-1500})

Mean values of water retention at -33 kPa ranged from 0.15 to 0.19 cm³ cm⁻³ (Table 1). The ANOVA for θ_{-33} measurements determined no significant difference between treatments or by depth, and no treatment $\times$ depth interaction effects. These results were somewhat expected since the PF was not hot enough or of long enough duration to alter the water retention properties of the soil (Alcañiz et al., 2018). While there were no significant differences among treatments, there was a slight decrease in θ_{-33} at both depths in the grassland postburn (Table 1), which aligns with a previous study that found that some PFs in grasslands can decrease the water retention of the soil (Badía et al., 2017). Across treatments, θ_{-33} values were not found to be significantly correlated with any other variable but were positively correlated to K_s , SOC, and NAG enzyme activity. Our findings align with prior studies that demonstrate that soil water retention is positively related to the soil's hydraulic conductivity (Touma, 2009), SOC content (Vereecken, 1995; Wagner et al., 2001), and NAG enzyme activity (Geisseler et al., 2011).

Mean values of volumetric water content at -1500 kPa ranged from 0.03 to 0.04 cm³ cm⁻³ between treatments (Table 1), with individual samples having a range of 0.02–0.05 cm³ cm⁻³ (data not shown). Analysis showed that mean

θ_{-1500} values decreased slightly at the grassland site at both evaluated depths after PF (Table 1, obscured due to significant figures), though no statistically significant differences were found between treatments or between depths, and there was no treatment $\times$ depth interaction effect. As has been noted by Garcia-Braga et al. (2024), given that PFs tend to be of low intensity, low severity, and patchy in nature, their effects depend on the type and quantity of accumulated fuel, soil moisture, and the amount of time between the fire and soil sampling. Across all treatments, θ_{-1500} was significantly positively correlated with SOC ($r = 0.79$, $p = 0.002$), total N ($r = 0.65$, $p = 0.021$), and NAG ($r = 0.60$, $p = 0.038$). Soil organic carbon is an important factor in soil water retention (Rawls et al., 2003), and because SOC affects the soil water holding potential and is linked with total N and NAG activity, the correlations found between soil water holding potential and total N and NAG are reasonable (Sainju et al., 2022).

3.7 | Aggregate stability

Differences in aggregate stability were not statistically significant between treatments or between depths sampled, and there were no treatment $\times$ depth interaction effects (Table 1). Despite a lack of statistical significance, aggregate stability in the grassland treatment increased at both measured depths after PF—by 8.7% at the 0- to 5-cm depth and by 7.1% at the 5- to 10-cm depth. This is similar to findings by Girona-García, Ortiz-Perpiñá, et al. (2018) who reported an increase in aggregate stability immediately after a PF; however, their aggregate size (0.25–0.55 mm) and sample depth (0–1 cm) are different from those in the present study. Similar findings (i.e., no significant difference in aggregate stability pre- and post-prescribed burn) have also been reported in other studies (Jordán et al., 2011; Scharenbroch et al., 2012). The lack of significant differences may be due to the size range of aggregates (1–2 mm) used in the wet aggregate stability measurements, as prior work has shown that larger aggregates (2–8 mm), rather than smaller aggregates (1–2 mm), are most affected by fire (Thomaz, 2017). Additionally, PFs with low burn temperatures like the one considered here may not have a strong effect on aggregate stability, even when burns occur regularly over a long period of time (Scharenbroch et al., 2012). Further, there is a possibility that the overnight rainfall before the PF and second sample collection may have obscured any effects of the PF on aggregate stability, though we would expect that the rainfall would have acted to decrease the impact of the fire on aggregate stability, since additional water in the soil would typically act to buffer temperatures at the soil surface during the burn. Aggregate stability was not significantly correlated with any other variable (Figure 3).

3.8 | Soil organic carbon and nitrogen

The results of the SOC analysis showed a significant difference ($p < 0.001$) between depths in both treatments but no significant differences between treatments or treatment $\times$ depth effects (Table 1). Previous studies have reported a variety of effects on SOC after PF, including a decrease in SOC immediately after fire (Girona-García, Badía-Villas, et al., 2018) and no change in SOC after fire (Chen et al., 2020; Fonseca et al., 2017). Soil organic carbon is generally located in the top 30 cm of the soil, with the highest concentration occurring in the upper 0–5 cm (Lorenz & Lal, 2005; Olson & Al-Kaisi, 2015). The effect of fire, whether wild or prescribed, on soil organic matter is dependent on a variety of factors including, but not limited to, soil characteristics, soil moisture content, and the severity of the burn (González-Perez et al., 2004; Shakesby, 2011). The lack of significant differences among treatments before and after the PF is likely due to the high soil water content and relatively low intensity of the burn, resulting in little to no effect on SOC, since PFs do not always reach a temperature high enough to affect SOC (Fontúrbel et al., 2012; Wang et al., 2013).

Across treatments, SOC data were found to be significantly positively correlated with θ_{-1500} ($r = 0.79$, $p = 0.002$), total N ($r = 0.86$, $p = 0.000$), and BG ($r = 0.83$, $p = 0.001$) and NAG ($r = 0.84$, $p = 0.001$) enzyme activity; they were also found to be significantly negatively correlated with bulk density ($r = -0.73$, $p = 0.007$). The positive relationship observed with θ_{-1500} and the negative correlation with bulk density are reasonable and align with prior studies that indicate that an increase in SOC improves soil structure (therefore increasing K_s , Yazdanpanah et al., 2016), decreases bulk density (Chaudhari et al., 2013), and increases soil water retention (Blanco-Canqui & Benjamin, 2013). Similarly, the strong positive relationship between SOC and enzyme activities was expected, given that BG and NAG play roles in the C and coupled C/N cycles, respectively. Our findings align with prior studies, which show a strong positive relationship between SOC and enzyme activity (de Almeida et al., 2015).

Mean total N content in the soil ranged from 1.20 g kg⁻¹ at the 5- to 10-cm depth in the grassland preburn to 2.24 g kg⁻¹ at the 0- to 5-cm depth in the grassland postburn. Total N contents were significantly different between depths, but no significant differences were found between treatments, and no treatment $\times$ depth interaction effects were found (Table 1). Total N was lower at both depths prior to the burn than after the burn, and was higher in the 0- to 5-cm depth both pre- and postburn. These trends align with findings from previous studies, which show that soil N content tends to be greatest near the surface (E. M. Smith et al., 2022) and that increases in nitrogen content within the top 10 cm of the soil surface may

occur immediately following fire (DeLuca & Zouhar, 2000). Across treatments, total N was significantly positively correlated with SOC ($r = 0.86$, $p = 0.000$), θ_{-1500} ($r = 0.65$, $p = 0.021$), BG ($r = 0.90$, $p = 0.000$) and NAG ($r = 0.83$, $p = 0.001$) activities, and K_s (0.69 , $p = 0.014$); it was also significantly negatively correlated with bulk density ($r = -0.82$, $p = 0.001$).

3.9 | Soil enzyme activity

Potential soil BG and NAG enzyme activities were significantly greater at the 0- to 5-cm soil depth compared to the 5- to 10-cm soil depth in all treatments ($p = 0.01$ for BG, $p = 0.02$ for NAG), but no significant differences were seen between treatments, and no treatment $\times$ depth interaction effects were found (Table 1). At the lower depth, only the grassland post-burn treatment had measurable BG activity (29.75 ± 26.61 mg PNP kg soil⁻¹), though this activity was low compared to values at the soil surface. Previous studies have reported a short-term reduction in soil extracellular enzyme activity following PFs, attributing the reductions to enzyme denaturation (Goberna et al., 2012; López-Poma & Bautista, 2014; Rai et al., 2023), with BG having greater declines compared to other enzymes (López-Poma & Bautista, 2014; Rai et al., 2023). Other studies that observed no changes in BG activity also observed no changes in SOC or reported that soil edaphic properties, such as soil moisture, had a greater influence on BG activity compared to PF (Boerner et al., 2000; Fontúrbel et al., 2016; Rietl & Jackson, 2012).

The lack of an effect of PF on enzyme activities in our study could be due to several factors, including the rainfall event the evening prior to the burn or the low burn intensity. Either of these factors may have buffered temperature increases during the burn, preventing enzyme denaturation (Campbell et al., 1995; Stoof et al., 2011). Our results indicate no short-term impacts of PF on enzyme activity. As discussed above, BG and NAG activities were negatively correlated with bulk density, positively correlated with SOC and N, BG positively correlated with K_s , and NAG with θ_{-1500} .

3.10 | LIMITATIONS

We acknowledge that there are several limiting factors that may impact our results. The first is the number of samples collected in the present study. Due to funding, time, and equipment limitations, it was only possible to collect samples from three locations (three samples from 0 to 5 cm and three samples from 5 to 10 cm, six samples in total) before and after the burn (12 samples total). We recognize that additional samples collected and analyzed under similar circumstances

could increase confidence in our results. Second, the rainfall event that occurred between soil samplings may have impacted our results. However, given that the soils at the site were already quite wet during the first sampling prior to the rainfall event and that the vegetation at the site was dry at the time of the burn, we believe the impact of the rainfall event on the parameters measured here was likely minimal. Finally, the impact of the study site's two historical PFs on the current results is unknown. Since soil samples were collected only before and after this third PF, there is no way for us to speak to the potential impacts of prior PFs on our results.

4 | CONCLUSION

This study sought to determine whether low-intensity PF had any significant effects on soil physical and chemical properties and enzymatic activity. We conducted soil sampling before and after a prescribed burn. While significant differences were found between depths for six soil properties—bulk density, saturated hydraulic conductivity, soil organic carbon content, total nitrogen, and BG and NAG enzyme activities—no significant differences were found in any soil properties when comparing samples pre- and postburn. The lack of significant differences, particularly in soil chemical properties and enzyme activities, before and after PF was unexpected and disproved our hypotheses that these properties would experience change. Our results suggest that low-intensity PF does not have any significant immediate effects on soil physical, chemical, or enzymatic properties at the Post Oak Savanna study site. This indicates that low-intensity PFs in this region are unlikely to significantly affect soil properties and functions. Given the short-term nature of this study, further research should be conducted within this ecoregion to determine whether long-term PF will affect soil properties and enzymatic activities.

AUTHOR CONTRIBUTIONS

Kathryn L. Watson: Conceptualization; data curation; formal analysis; investigation; methodology; visualization; writing—original draft; writing—review and editing. **Jordan E. Gignac:** Conceptualization; data curation; formal analysis; investigation; methodology; writing—original draft; writing—review and editing. **Alejandro Atenas Navarrete:** Conceptualization; data curation; formal analysis; investigation; methodology; visualization; writing—original draft; writing—review and editing. **Leila Joharzadeh:** Conceptualization; data curation; formal analysis; investigation; methodology; writing—original draft; writing—review and editing. **Bismark Osei:** Conceptualization; data curation; formal analysis; investigation; methodology; writing—original

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data from this study are available upon request to the corresponding author.

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